

Total No. of Questions : 6]

SEAT No. :

P24

Oct./TE/Insem.-22

[Total No. of Pages : 2

T.E. (Information Technology)

COMPUTER NETWORK TECHNOLOGY

(2012 Pattern) (Semester - I) (314441)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Explain count to infinity problem in detail. How to resolve it? **[6]**

b) Differentiate between packet switching and virtual circuit switching. **[4]**

OR

Q2) a) Write the subnet, broadcast address, and valid host range for 192.168.100.157/27. **[6]**

b) Why do OSPF messages propagate faster than RIP messages? **[4]**

Q3) a) Describe working of Leaky bucket and Token bucket algorithm with reference to CBR, VBR and bursty traffic. **[6]**

b) What features of UDP makes it connection-less protocol? Explain in detail. **[4]**

OR

Q4) a) What is socket? Explain various socket primitives used in connection oriented client server interaction. **[6]**

b) Explain how flowcontrol is implemented in TCP. **[4]**

P.T.O.

Q5) a) Explain the HTTP transaction between the client and server with a suitable diagram. Differentiate between persistent and non-persistent HTTP connection. **[6]**

b) Explain in brief SNMP protocol. **[4]**

OR

Q6) a) What is the purpose of DNS? Explain the following terms with reference to DNS **[6]**

i) Label.

ii) FQDN/PQDN.

iii) Domain.

iv) Zone.

b) Why FTP uses two different ports for its operation? Explain any 4 commands in FTP. **[4]**
